QUESTIONS

These questions must be answered individually. We believe it is possible to answer these questions adequately using around 600 words in total. You can use code snippets to illustrate your point.

1. What is the difference between a class and an object?

   A class is a recipe for an object. And vice versa the object is an instance of a class.

2. What is a design pattern? Give an example from your Mayhem implementation.

   A design pattern is a common way of solving a problem. My code used a Template pattern which is an implementation that is useful for removing duplicate code. It is designed for situations with similar tasks that share most steps. An example from my code would be the instances I make in the game class initializing. Here I call to several other classes and later in the game loop call to their specific methods to update information.

3. What is the difference between a has-a and an is-a relationship?

   A has-a relationship is when a class has an instance of another class implemented. A is-a relationship is when a class inherits information from another class. The difference being that when a class is inherited, the "child" or the recipient receives all of the information from the "parent" or the sender. While when a class instance is implemented, the information it initializes is sent from the class instantiating it. So, the class is not connected the same way, but selected information is shared.

4. What is encapsulation? How is encapsulation handled in Python?

   Encapsulation is to hide data and data functions in the object. Encapsulation is generally not used in python, but an example would be the initializer definition for an object. Using a single or double underscore makes it so that the when the class is instantiated, the method is run without a direct call.

5. What is polymorphism? Give examples of polymorphism from the precode and the Mayhem implementation.

   Polymorphism is when one or more entity type(s) can solve the same problem(s) in different ways. An example from the precode could be the mathematical definitions. These definitions take two variables and does a mathematical function (like add or substract e.g) with the same logic as the math functions but in different ways. They handle vectors like they were integers. An example from my game implementation would be the spaceship class. It is instantiated twice with different information, but goes through the same function and makes the same changes.